1.

a.

First of all we will notice that there are $2^{10}$ different states as there are 10 different transition systems with 2 states each. Lets represent the states as $< S_T, S_1,..., S_9 >$. As for reachable states, notice that $< S_0, S_1, S_1,..., S_1 > \ \ and \ \ < S_1, S_0, S_{0,},..., S_0 >$ are unreachable states as when moving in $TRN$ one ‘step’ forward we have to move in one of the other steps forward, and also we can't move in every other system forward to $S_1$ without moving at all with $TRN$. That means we have $2^{10} - 2$ reachable states which means 1022.

b.

Notice its tic tac toe without rules, a different run is based on the different labels we collected in the entire run, the number of options to fill a tic tac toe board with X and O is:

$9! \ = \ 362,880$

c.

We will make 8 new transition systems, one for every column, row and diagonal. Two of each, one for X and one for O, every one of these TS we will make 4 states, one for empty, one for when one of them is taken, one for when two spots are taken and one for when the line is completed, when reaching this state its game over. When we interweave the systems we will receive a tic tac toe game! nice!

notice there are 8 rows columns and diagonals total, so we added 16 transition systems, each has 4 states, now we have overall $1022 \cdot 4^{16}$ states overall.

The reachable states question is quite complicated because when for example a row is taken by X it can't be taken by O so it does rule out many options.

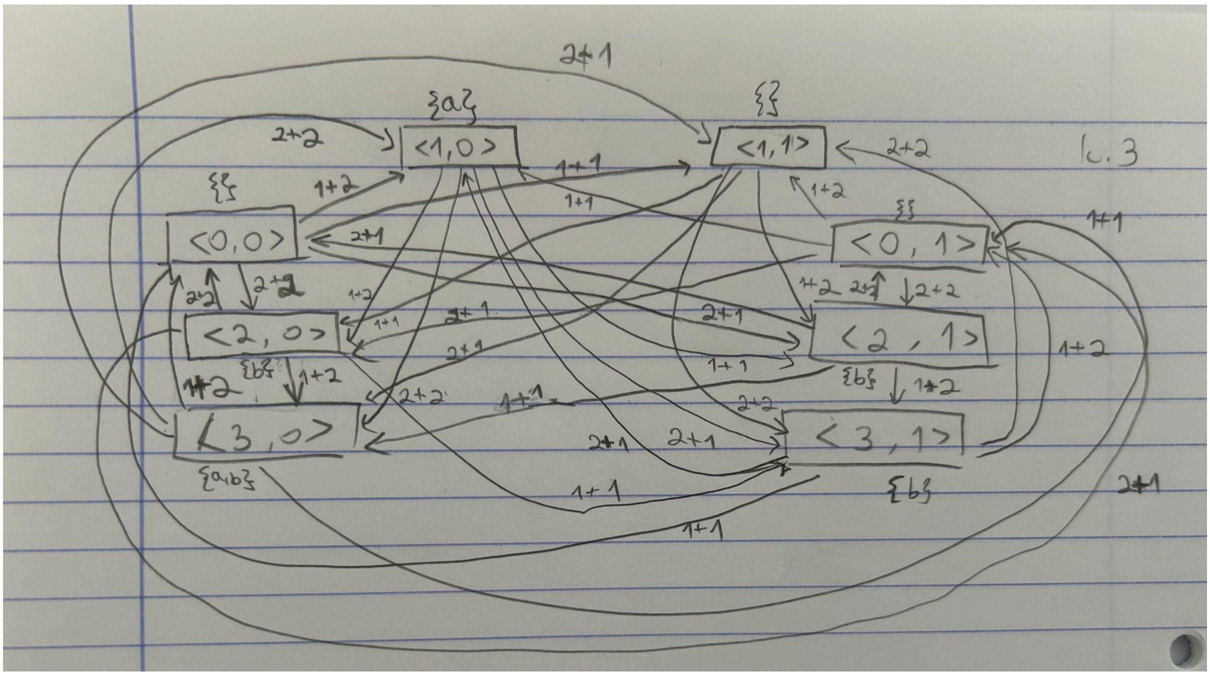
There are $3^9 = 19683$ different runs of tic tac toe.

2.

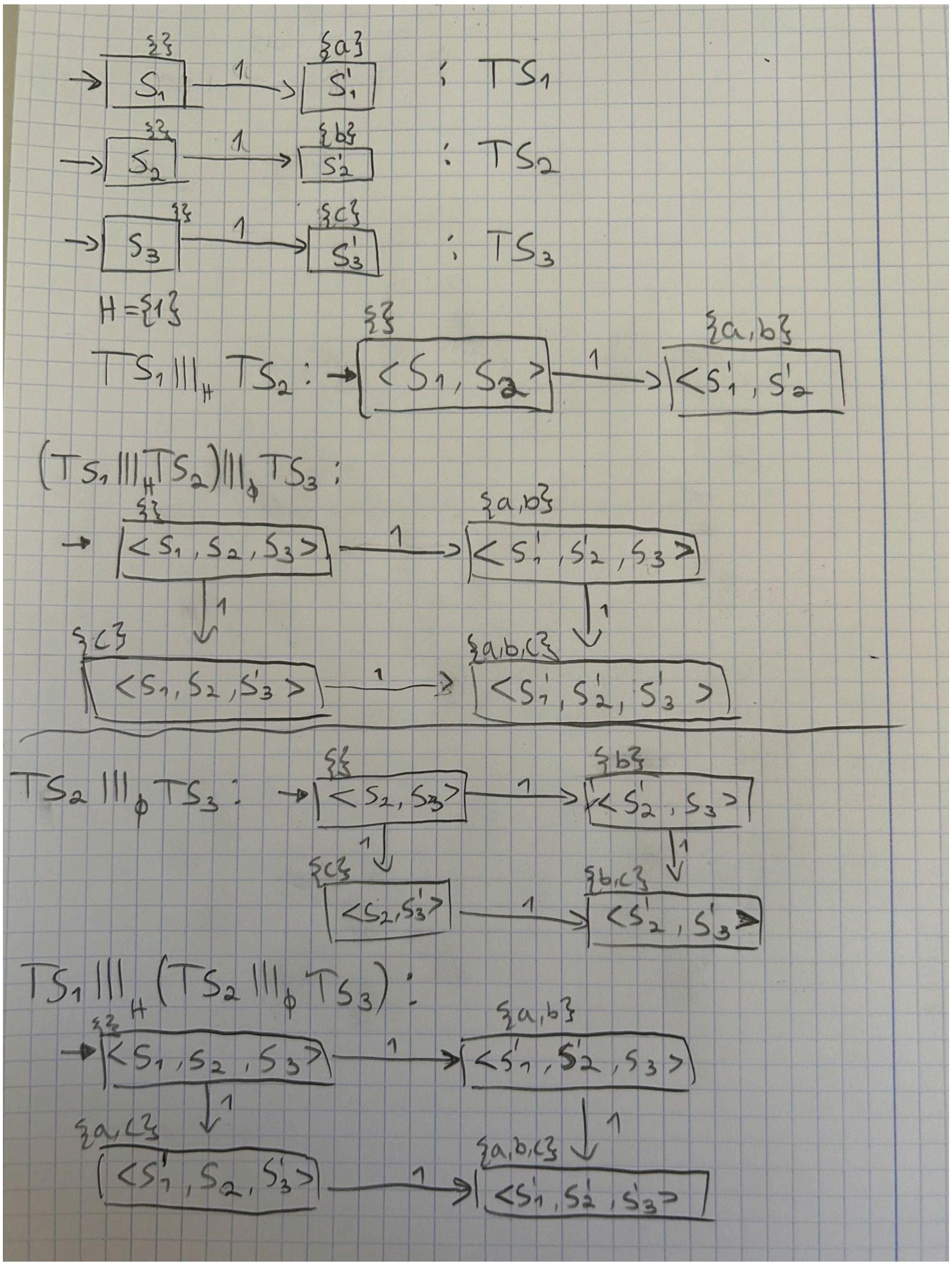
Let's have a look at $PG_0$, notice that it never reaches cs. By interweaving the systems we don't change the behavior of $PG_0$ and we will never reach a ‘Bad’ state.

3.

a.

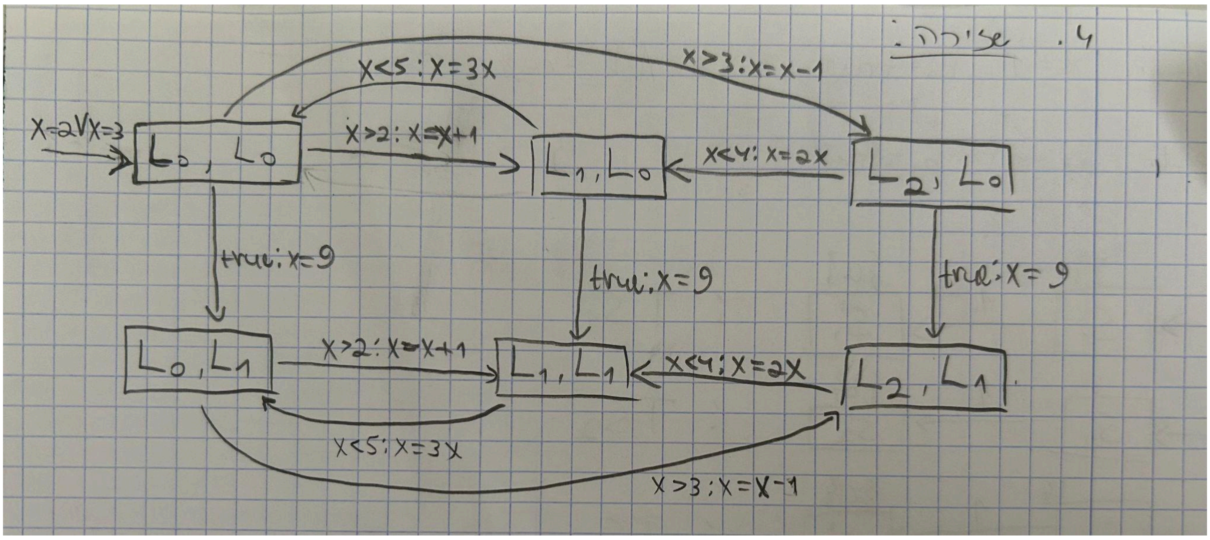


b.
Here is the Example:



As you can see $(TS_1 |||_H TS_2) |||_\phi TS_3 \neq TS_1 |||_H (TS_2 |||_\phi TS_3)$.

4.
Interweaving:




5.
a.
The statement is false.Here is a contradicting example: